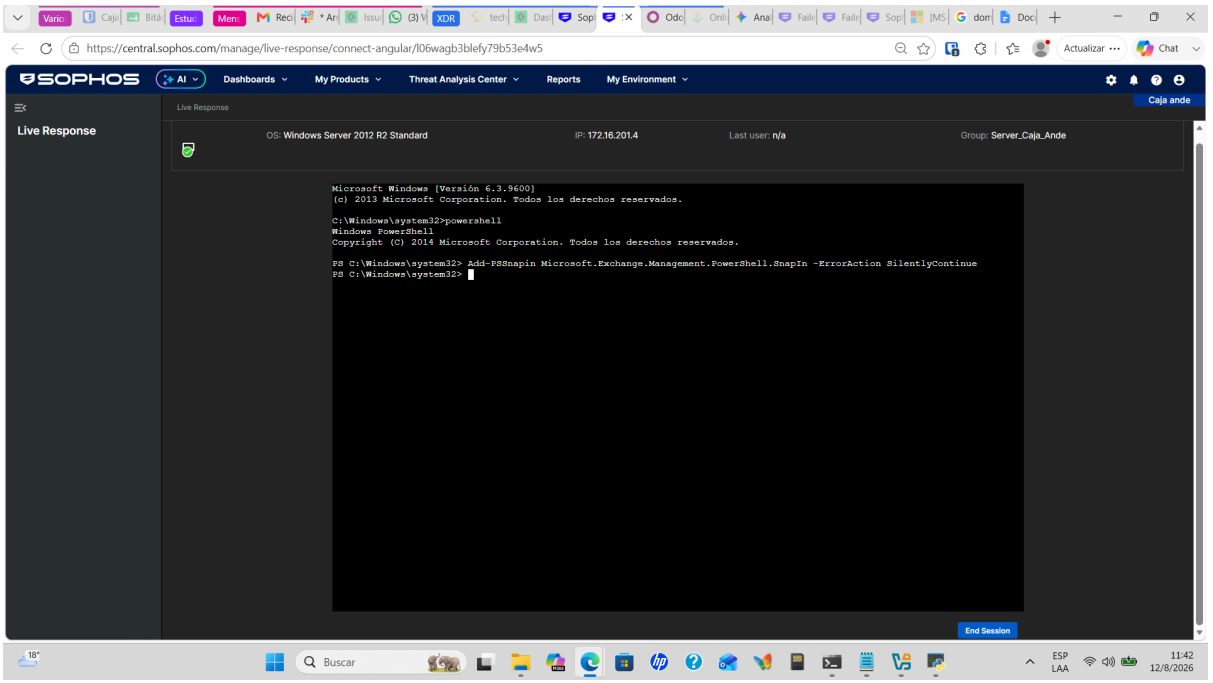
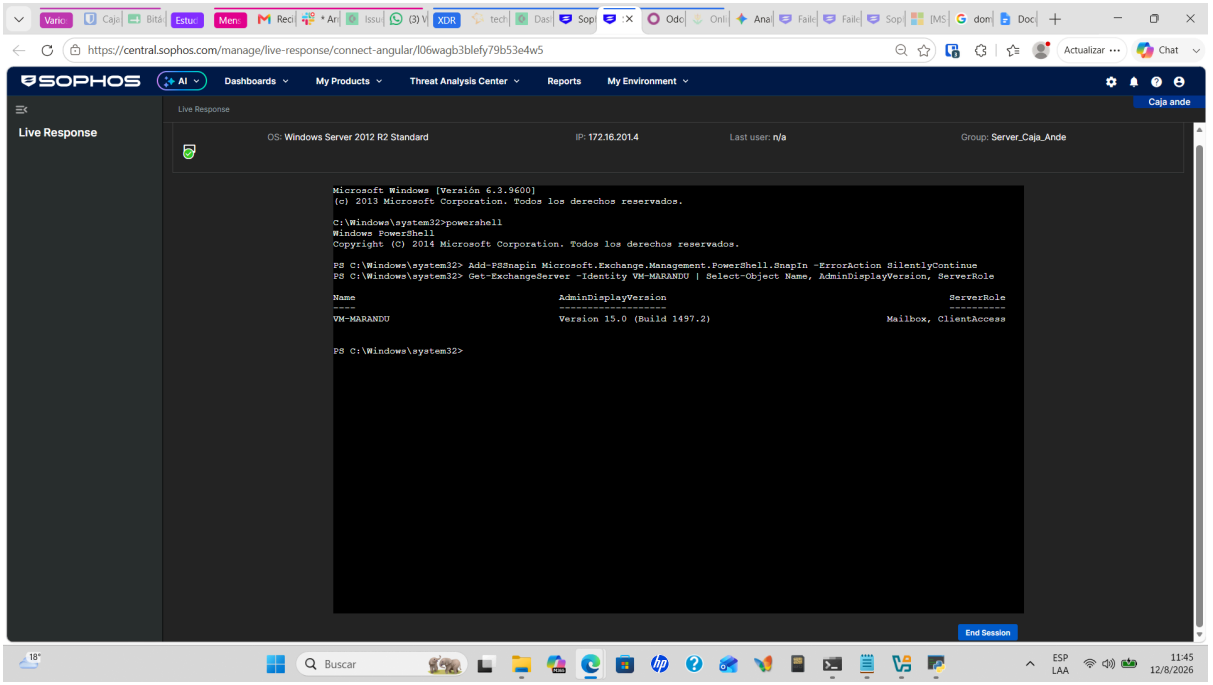
0. Inicialización Obligatoria en PowerShell (Carga del Módulo de Exchange)

```
None
Add-PSSnapin Microsoft.Exchange.Management.PowerShell.SnapIn -ErrorAction
SilentlyContinue
```



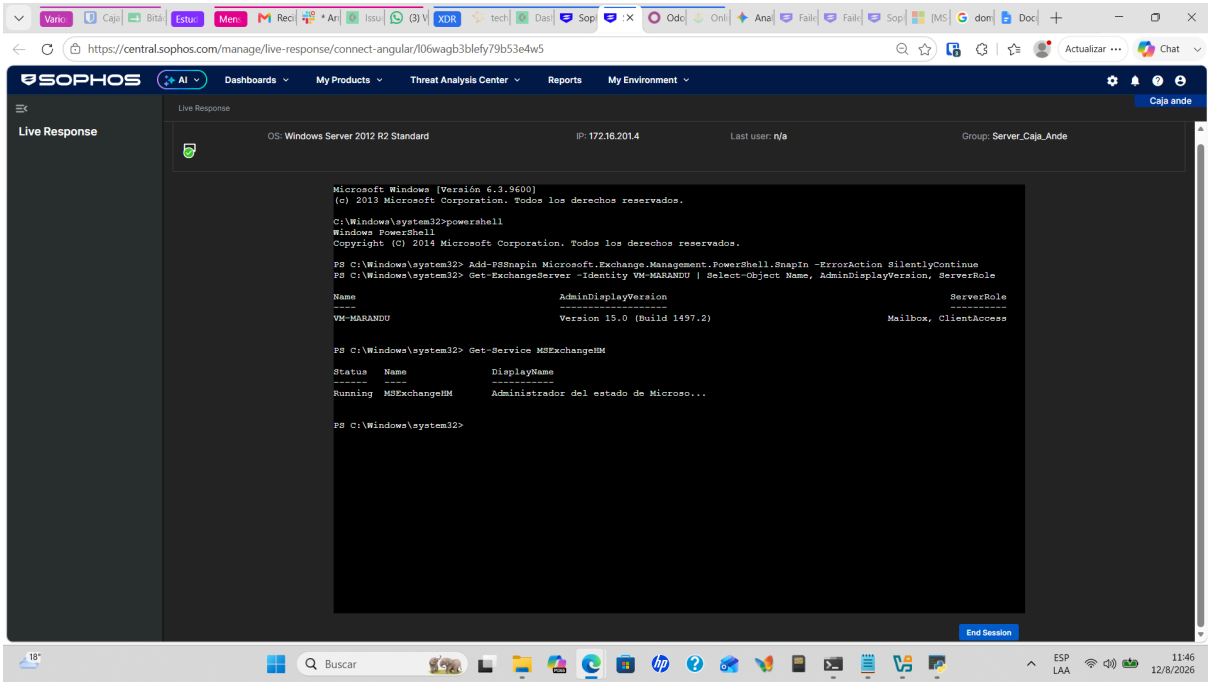
1. Identificación del Servidor, Roles y Software
   - Consultar versión build exacta y roles de Exchange (Mailbox/ClientAccess)

```
None
Get-ExchangeServer -Identity VM-MARANDU | Select-Object Name,
AdminDisplayVersion, ServerRole
```



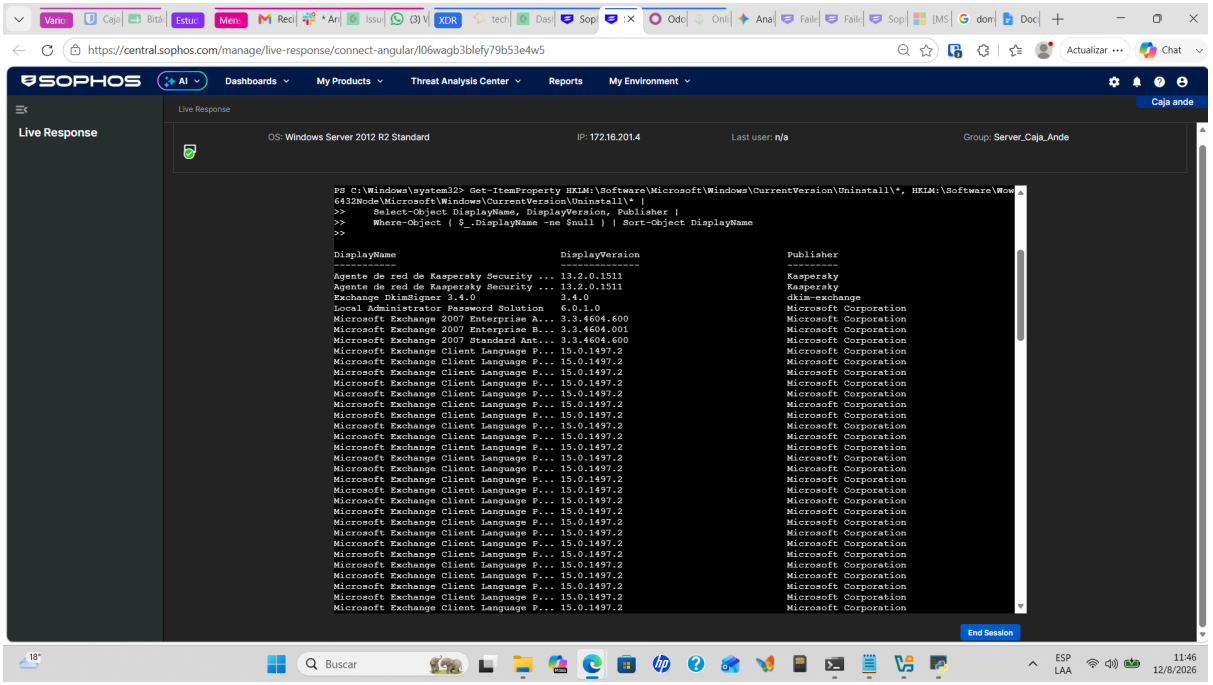
- Verificar estado del servicio Health Manager (MSExchangeHM)

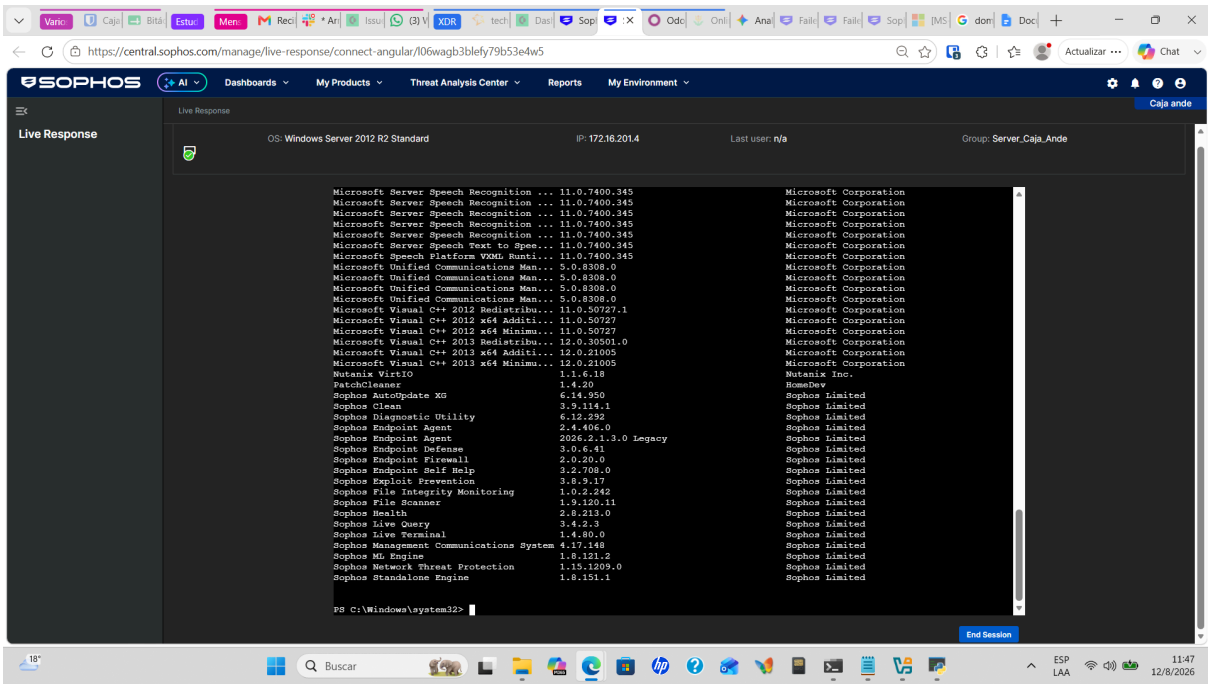
```
None
Get-Service MSExchangeHM
```



- Inventario completo de aplicaciones instaladas en el registro (32 y 64 bits)

```
None
Get-ItemProperty HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall\*,
HKLM:\Software\Wow6432Node\Microsoft\Windows\CurrentVersion\Uninstall\* |
    Select-Object DisplayName, DisplayVersion, Publisher |
    Where-Object { $_.DisplayName -ne $null } | Sort-Object DisplayName
```

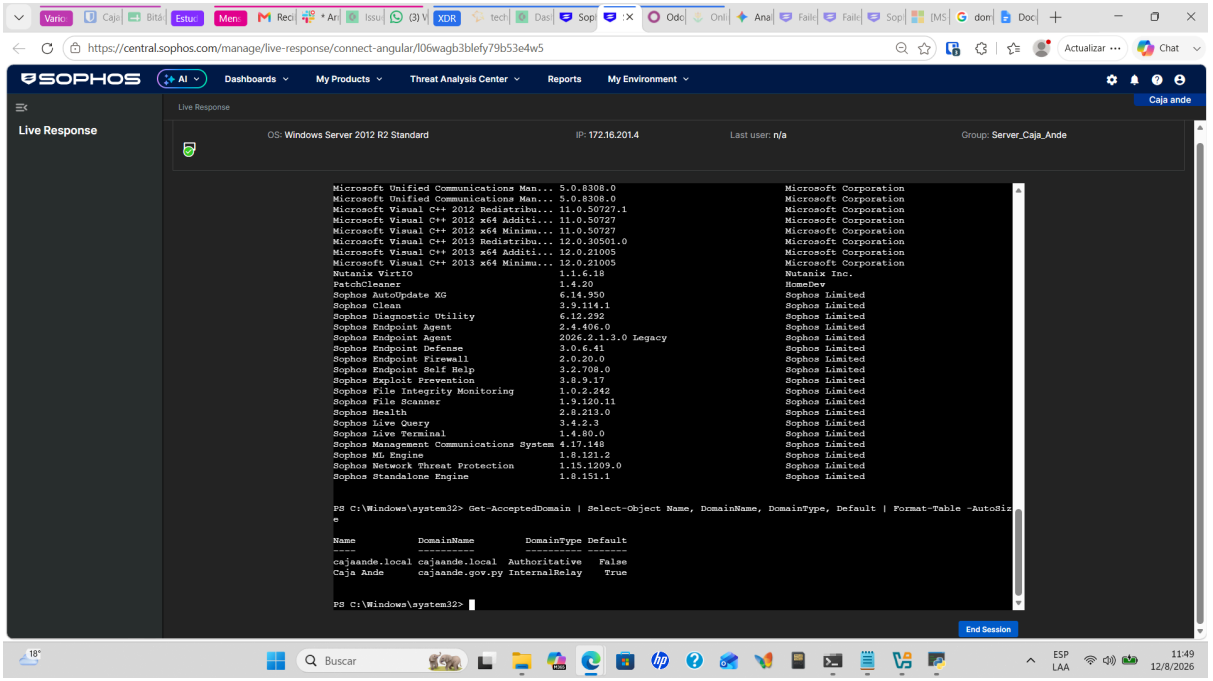




2. Diagnóstico de Configuración de Dominios y Políticas de Correo
   - Consultar dominios aceptados (para verificar Authoritative vs InternalRelay)

None

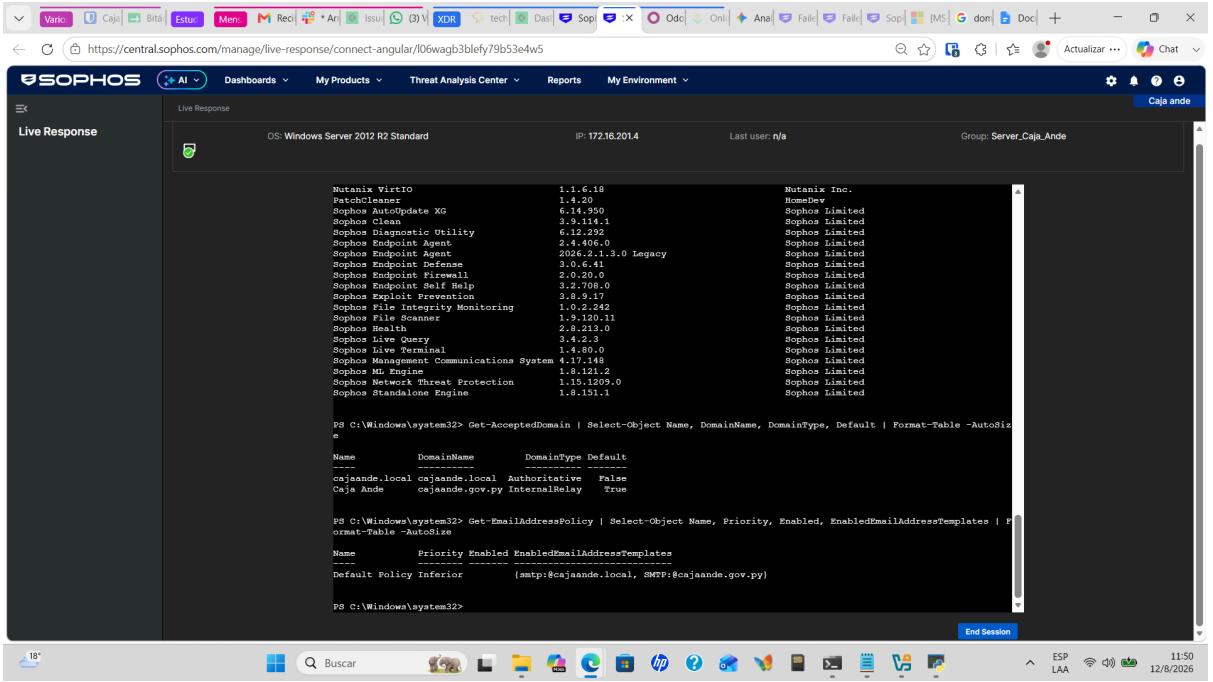
```
Get-AcceptedDomain | Select-Object Name, DomainName, DomainType, Default | Format-Table -AutoSize
```



- Consultar política de direcciones de correo (dirección SMTP primaria vs secundarios)

None

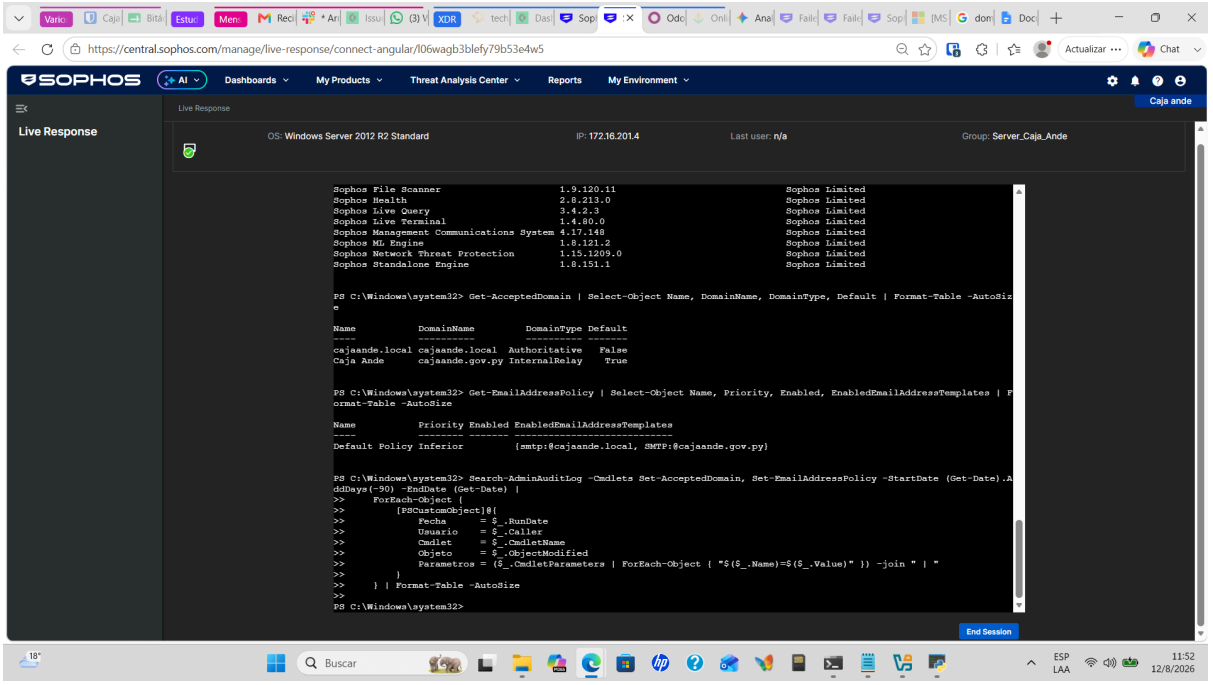
```
Get-EmailAddressPolicy | Select-Object Name, Priority, Enabled, EnabledEmailAddressTemplates | Format-Table -AutoSize
```



3. Auditoría de Cambios y Registro de Eventos (Últimos 90 Días)
   - Consulta a AdminAuditLog de Exchange (Busca ejecuciones de Set-AcceptedDomain)

None

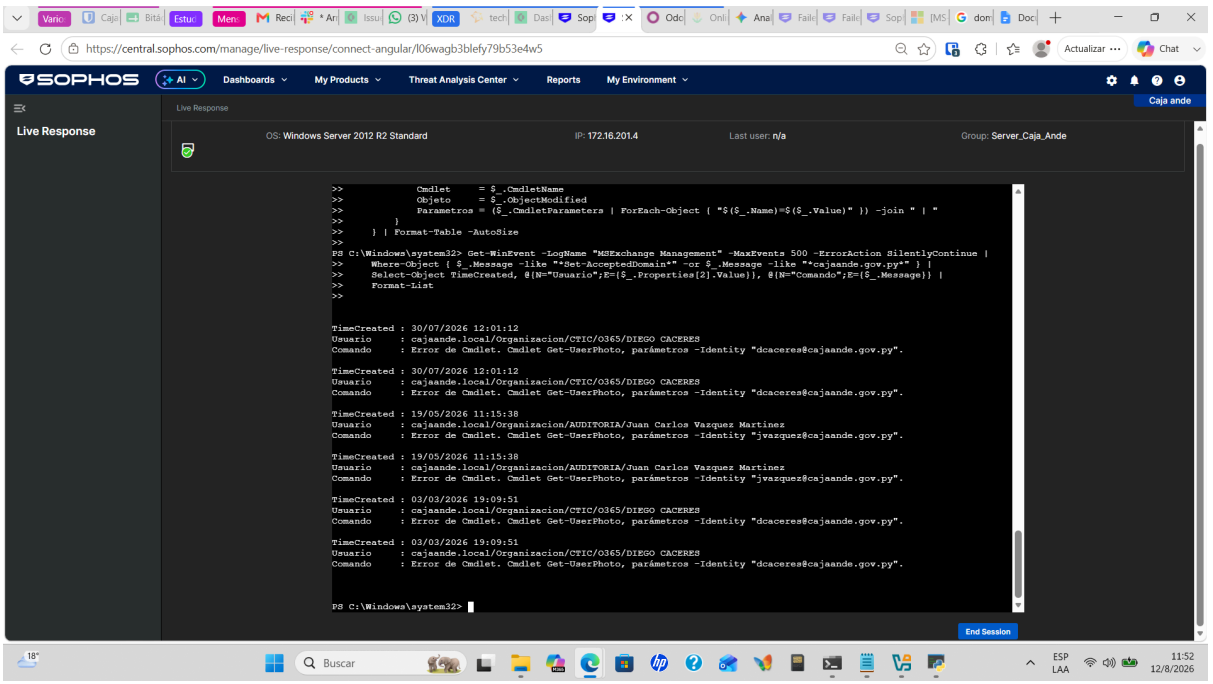
```
Search-AdminAuditLog -Cmdlets Set-AcceptedDomain, Set-EmailAddressPolicy
-StartDate (Get-Date).AddDays(-90) -EndDate (Get-Date) |
    ForEach-Object {
        [PSCustomObject]@{
            Fecha      = $_.RunDate
            Usuario    = $_.Caller
            Cmdlet     = $_.CmdletName
            Objeto     = $_.ObjectModified
            Parametros = ($_.CmdletParameters | ForEach-Object {
"$($_.Name)=$($_.Value)" }) -join " | "
        }
    } | Format-Table -AutoSize
```



   - Consulta al Event Viewer de Windows (Canal MSExchange Management)

None

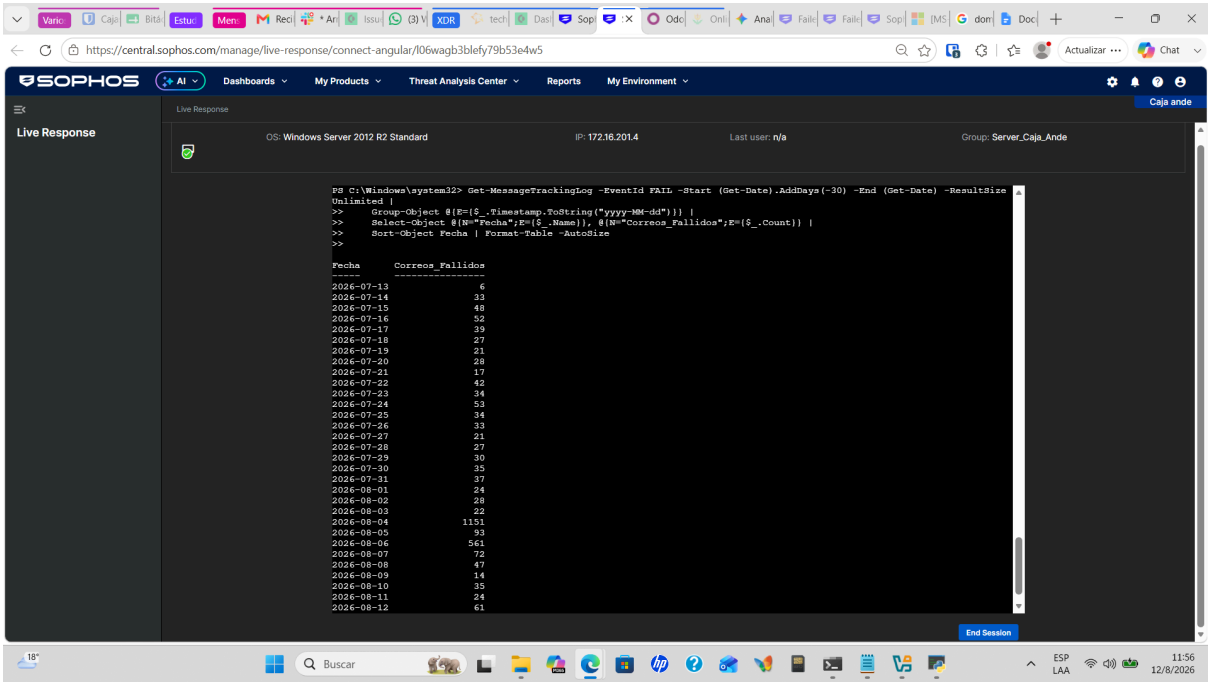
```
Get-WinEvent -LogName "MSExchange Management" -MaxEvents 500 -ErrorAction
SilentlyContinue |
    Where-Object { $_.Message -like "*Set-AcceptedDomain*" -or $_.Message -like
"*cajaande.gov.py*" } |
    Select-Object TimeCreated, @{N="Usuario";E={$_.Properties[2].Value}},
@{N="Comando";E={$_.Message}} |
    Format-List
```



4. Medición del Impacto y Cuantificación de Rebotes (Message Tracking Log)
   - Conteo diario de fallos de entrega (FAIL) en los últimos 30 días

None

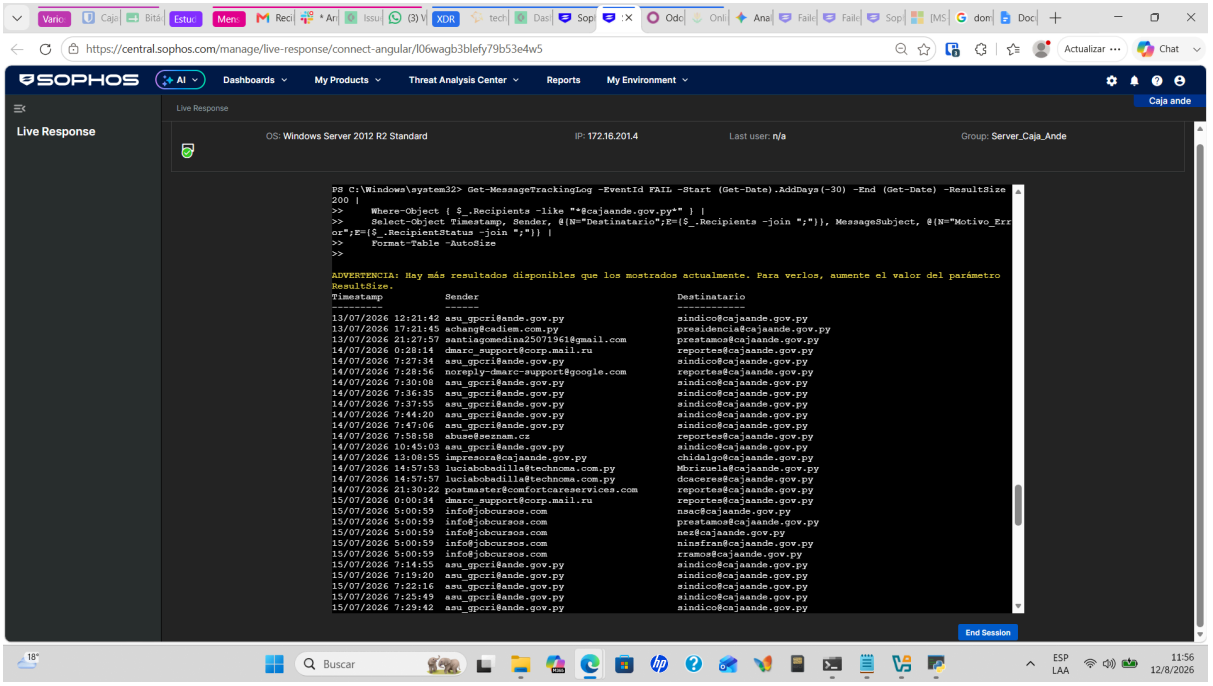
```
Get-MessageTrackingLog -EventId FAIL -Start (Get-Date).AddDays(-30) -End
(Get-Date) -ResultSize Unlimited |
    Group-Object @{E={$_.Timestamp.ToString("yyyy-MM-dd")}} |
    Select-Object @{N="Fecha";E={$_.Name}},
@{N="Correos_Fallidos";E={$_.Count}} |
    Sort-Object Fecha | Format-Table -AutoSize
```

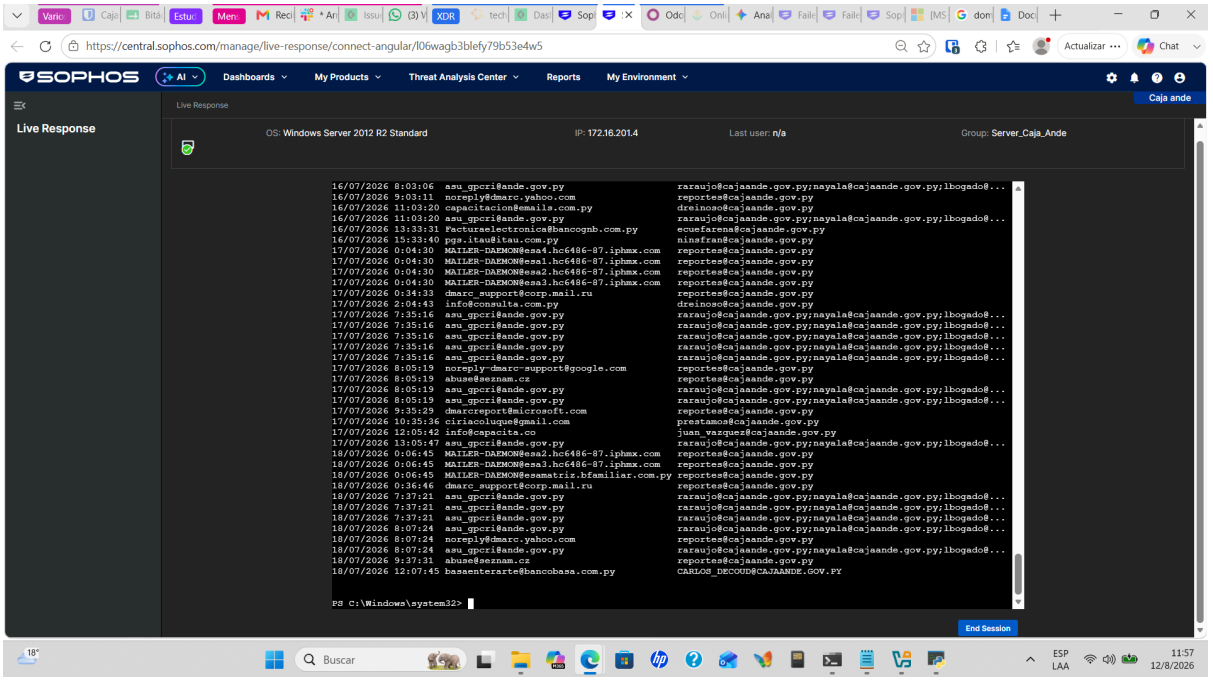


- Muestreo de remitentes externos, destinatarios y motivos de rebote hacia @cajaande.gov.py

None

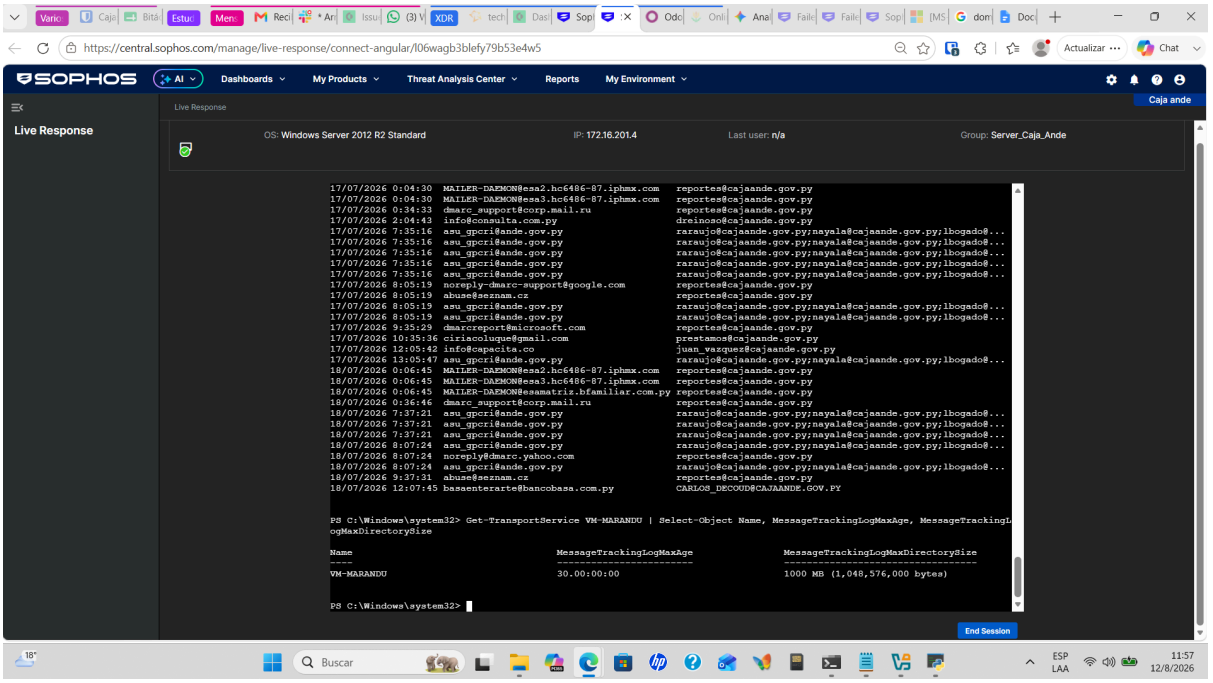
```
Get-MessageTrackingLog -EventId FAIL -Start (Get-Date).AddDays(-30) -End
(Get-Date) -ResultSize 200 |
    Where-Object { $_.Recipients -like "*@cajaande.gov.py*" } |
    Select-Object Timestamp, Sender, @{N="Destinatario";E={$_.Recipients -join
";"}}, MessageSubject, @{N="Motivo_Error";E={$_.RecipientStatus -join ";"}} |
    Format-Table -AutoSize
```





- Parámetros de retención del log de transporte en el servidor

```
Get-TransportService VM-MARANDU | Select-Object Name, MessageTrackingLogMaxAge, MessageTrackingLogMaxDirectorySize
```



REMEDIACIÓN

```
Set-AcceptedDomain -Identity "Caja Ande" -DomainType Authoritative
```